

01

Determinant of the Matrix

What is a determinant of a matrix? How to calculate it?

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What is a Determinant?

1. The determinant of a matrix is a **scalar value** (a single number) that is computed from a **square matrix**.
2. It is usually denoted by $\det(A)$ or $|A|$
3. The determinant is used to:
 - a. Determine whether a matrix is invertible:

$$\det(A) = 0 \begin{cases} \text{true} & \text{Matrix A has no inverse} \\ \text{false} & \text{Matrix A has an inverse} \end{cases}$$

- b. Decide whether a system of linear equations has a unique solution;
- c. Describe how a linear transformation scales area or volume.

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Basic Solution

1. The use of determinants arose from the recognition of **special patterns** that occur in the solutions of systems of linear equations.
2. For example, if there is an SLE as below:

$$2x_1 + x_2 = 4$$

$$-x_1 + 2x_2 = 3$$

can be denoted as follows:

$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

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Basic Solution

$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

can be shown to be:

$$x_1 = \frac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{21}a_{12}} \quad \text{and} \quad x_2 = \frac{b_2a_{11} - b_1a_{21}}{a_{11}a_{22} - a_{21}a_{12}}$$

Provided that $a_{11}a_{22} - a_{21}a_{12} \neq 0$

Both fractions have the same denominator, called the determinant of the **coefficient matrix**.

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Inverse of a Matrix

1. The inverse of a matrix is denoted by A^{-1}
2. As we know, in the common mathematics $x \cdot x^{-1} = 1$ applies, then in matrices $A \cdot A^{-1} = I$ applies.
3. We will discuss more about inverse of a matrix in another meeting.

Determinant of a 2x2 Matrix

1. For a 2x2 matrix, we can get the determinant value by using the following formula:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow |A| = ad - bc$$

2. $ad - bc$ also used when finding the inverse of a 2x2 matrix, with the formula:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

So the formula for finding the inverse of a matrix can also be written as:

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example: 2x2 Matrix

If $A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$ then $|A| = ad - bc = 1 \cdot 0 - 2 \cdot (-1) = 2$

$$\text{and } A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\text{where } A \cdot A^{-1} = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Example: 2x2 Matrix

Determine the determinant value of the following matrix:

$$1. \quad A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$$

$$2. \quad B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

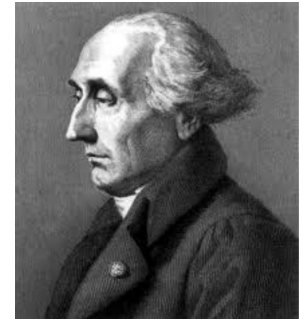
$$3. \quad C = \begin{bmatrix} 5 & -4 \\ 2 & 3 \end{bmatrix}$$

Remarks

1. The value of the determinant of a matrix **can be** positive, zero, or negative.
2. The determinant of a matrix of order 1 is defined simply as the entry of the matrix.
For instance, if $A = [-2]$ then $|A| = -2$
3. To define the determinant of a matrix of order **higher than 2**, it is convenient to use the notions of **minors** and **cofactors**.

Determinant of a 3x3 Matrix

1. One way to solve the determinant problem of a **3x3 matrix** is by using the **Sarrus method**.
2. The Sarrus method is considered the easiest method to use to calculate the determinant value of a matrix.
3. This method is **based on** the method of calculating the determinant of a **2x2 matrix**.
4. But unfortunately the Sarrus method **cannot** be used for matrices that have an order of more than 3.
5. Order more than 3 → expansion → Sarrus method.



Pierre Frédéric Sarrus
1798 - 1861

The Formula

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Rightarrow |A| = (a \cdot e \cdot i) + (b \cdot f \cdot g) + (c \cdot d \cdot h) - (c \cdot e \cdot g) - (a \cdot f \cdot h) - (b \cdot d \cdot i)$$

Example:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 1 \\ 2 & 3 & -1 \end{bmatrix} \Rightarrow |A| = (1 \cdot 0 \cdot (-1)) + (2 \cdot 1 \cdot 2) + (0 \cdot (-1) \cdot 3) - (0 \cdot 0 \cdot 2) - (1 \cdot 1 \cdot 3) - (2 \cdot (-1) \cdot (-1))$$

$$= 0 + 4 + 0 - 0 - 3 - 2$$

$$= -1$$

How to?

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{array}{l} \text{+} \\ \text{+} \\ \text{+} \\ \text{-} \\ \text{-} \\ \text{-} \end{array}$$



1. Copy the **first 2 column** as column 4 and 5.
2. Multiply the numbers in each of the three diagonals that run **from the top-left to the bottom-right**. These diagonals have **positive sign**.
3. Multiply the numbers in each of the three diagonals that run **from the top-right to the bottom-left**. These diagonals have **negative sign**.

Example

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix} \Rightarrow |A| =$$



02 Minor and Cofactor

Find out what the role of minor and cofactors is in determinants.

Minor of a Matrix

1. **Each element** of a square matrix has a minor.
2. For a matrix A , the minor of an element a_{ij} (the entry in row i , column j) is **the determinant of the submatrix obtained by deleting row i and column j from A**
3. This minor is usually denoted by M_{ij}
4. Thus then:
 - a. $A_{1 \times 1}$ matrix has no minor matrices
 - b. $A_{2 \times 2}$ matrix has a minor matrix with 4 elements M_{11} , M_{12} , M_{21} , and M_{22}
 - c. $A_{3 \times 3}$ matrix has a minor matrix with 9 elements M_{11} , M_{12} , M_{13} , M_{21} , M_{22} , M_{23} , M_{31} , M_{32} , and M_{33}
 - d. and so on

Example: 2x2 Matrix

If matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

then $M_{11} = \begin{bmatrix} \cancel{1} & \cancel{2} \\ \cancel{3} & 4 \end{bmatrix} = 4$

$M_{12} = \begin{bmatrix} \cancel{1} & \cancel{2} \\ 3 & \cancel{4} \end{bmatrix} = 3$

$M_{21} = \begin{bmatrix} \cancel{1} & 2 \\ \cancel{3} & \cancel{4} \end{bmatrix} = 2$

$M_{22} = \begin{bmatrix} 1 & \cancel{2} \\ 3 & \cancel{4} \end{bmatrix} = 1$

minor of the matrix A is $\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$

Example: 3x3 Matrix

$$B = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 2 & -1 & 1 \end{bmatrix}$$



Cofactor of a Matrix

1. Similar to minors, each element of a square matrix has a cofactor.
2. Cofactor of the element a_{ij} is defined as $C_{ij} = (-1)^{i+j} M_{ij}$

where:

M_{ij} is the **minor** of a_{ij}

$(-1)^{i+j}$ is the **sign factor**

if $i + j$ is **even**, then the sign factor is **+**

if $i + j$ is **odd**, then the sign factor is **-**

therefore:

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots \\ C_{21} & C_{22} & C_{23} & \cdots \\ C_{31} & C_{32} & C_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \rightarrow \begin{bmatrix} +M_{11} & -M_{12} & +M_{13} & \cdots \\ -M_{21} & +M_{22} & -M_{23} & \cdots \\ +M_{31} & -M_{32} & +M_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Example: 2x2 Matrix

If matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

then $M_{11} = \begin{bmatrix} \cancel{1} & \cancel{2} \\ \cancel{3} & 4 \end{bmatrix} = 4$

$M_{12} = \begin{bmatrix} \cancel{1} & \cancel{2} \\ 3 & \cancel{4} \end{bmatrix} = 3$

$M_{21} = \begin{bmatrix} \cancel{1} & 2 \\ \cancel{3} & \cancel{4} \end{bmatrix} = 2$

$M_{22} = \begin{bmatrix} 1 & \cancel{2} \\ 3 & \cancel{4} \end{bmatrix} = 1$

cofactor of the matrix A is $\begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$

Example: 3x3 Matrix

$$B = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 3 & 2 \\ 2 & -1 & -1 \end{bmatrix}$$



03 Laplace Expansion Method

How to calculate the determinant using the Laplace expansion method?

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What is It?

1. Used to calculate the determinant value of a matrix that has an order **greater than 3**.
2. This method was introduced by **Pierre-Simon Laplace**, a French engineer, who was an expert in mathematics, statistics, physics, astronomy, and philosophy.
3. The determinant value of a matrix can be calculated by performing a **cofactor expansion**, **either on a particular row or column that can be chosen freely**.



Pierre-Simon Laplace
1749 - 1827

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The Formula

$$|A| = \begin{cases} \sum_{j=1}^n a_{ij}c_{ij} & \rightarrow \text{cofactor expansion along row } i \\ \sum_{i=1}^n a_{ij}c_{ij} & \rightarrow \text{cofactor expansion along column } j \end{cases}$$

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Example: 3x3 Matrix

$$A = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 0 & 1 \\ 4 & 2 & -2 \end{bmatrix} \xrightarrow{\text{cofactor expansion along row 1}} A = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 0 & 1 \\ 4 & 2 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 2 \\ -3 & 0 & 1 \\ 4 & 2 & -2 \end{bmatrix} \quad M_{11} = (0 \cdot (-2)) - (1 \cdot 2) = 0 - 2 = -2 \\ C_{11} = -2$$

$$\begin{bmatrix} 1 & -3 & 2 \\ -3 & 0 & 1 \\ 4 & 2 & -2 \end{bmatrix} \quad M_{12} = ((-3) \cdot (-2)) - (1 \cdot 4) = 6 - 4 = 2 \\ C_{12} = -2$$

$$\begin{bmatrix} 1 & -3 & 2 \\ -3 & 0 & 1 \\ 4 & 2 & -2 \end{bmatrix} \quad M_{13} = ((-3) \cdot 2) - (0 \cdot 4) = -6 - 0 = -6 \\ C_{13} = -6$$

$$\begin{aligned} |A| &= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} \\ &= 1 \cdot (-2) + (-3) \cdot (-2) + 2 \cdot (-6) \\ &= (-2) + 6 + (-12) \\ &= -8 \end{aligned}$$

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